AI Systems

& Their Impact

A Comprehensive Overview for the Next Generation of AI Practitioners

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What is an AI System?

Slide 1 of 12 — Foundations

Definition: An AI System is a computational entity that mimics human intelligence — it **Perceives → Thinks → Learns → Acts** to achieve goals in its environment.

1

PERCEPTION

Sensors, cameras, text, audio, images

2

REASONING

Logic, rules, algorithms, planning

3

LEARNING

ML, Deep Learning, reinforcement

4

ACTION

Robots, software, output, decisions

💡 *Expert Insight: Unlike traditional software, AI systems improve with experience — they don't just execute; they evolve.*

AI System Architecture: The 4-Layer Model

Slide 2 of 12 — System Design

1 **INPUT / PERCEPTION LAYER**

Raw data ingestion from sensors, datasets, cameras, microphones, text streams

2 **REASONING / DECISION LAYER**

Rules engines, neural networks, knowledge graphs, inference algorithms

3 **LEARNING / ADAPTATION LAYER**

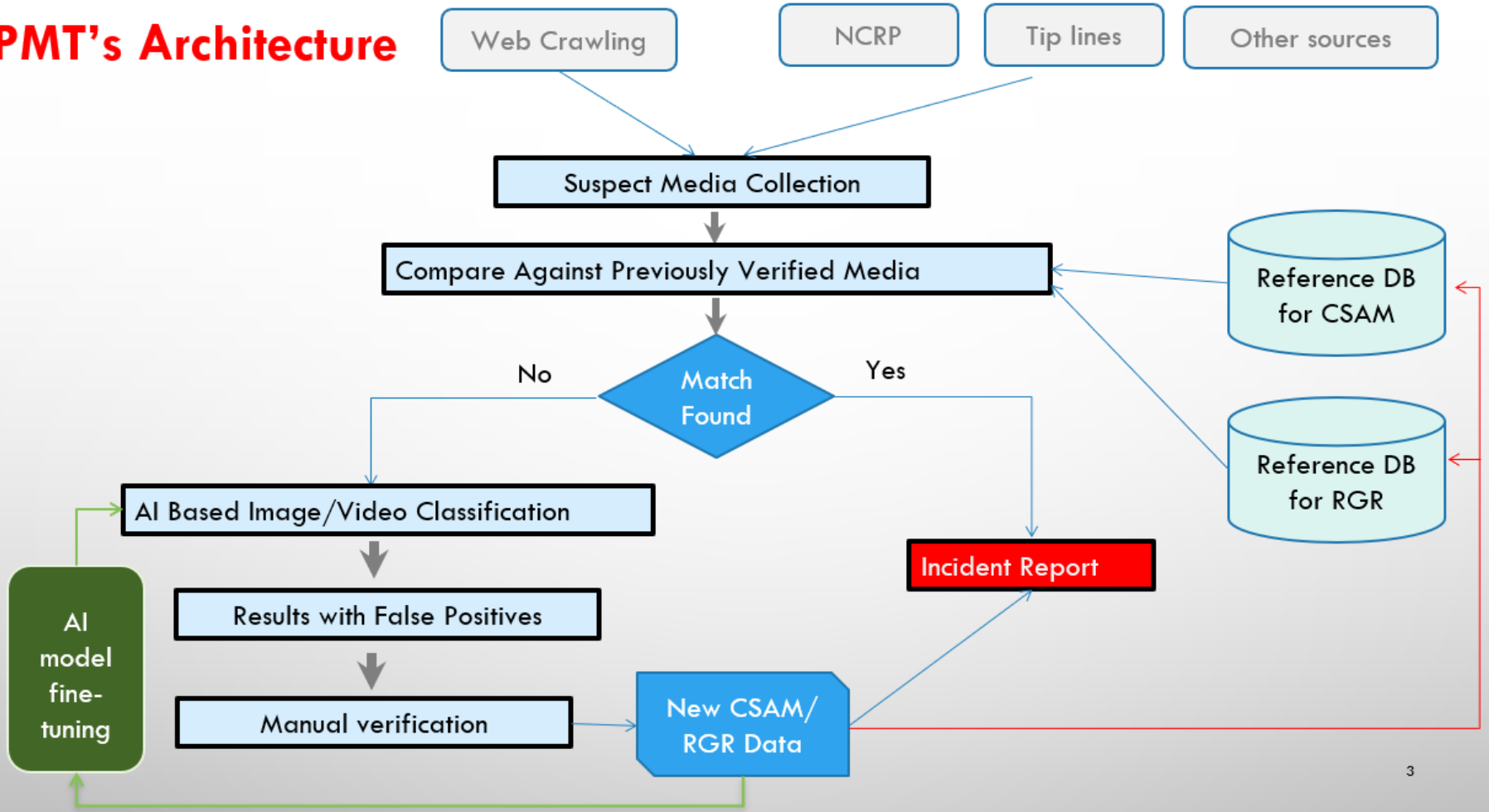
Model training, weight updates, reinforcement loops, transfer learning

4 **ACTION / OUTPUT LAYER**

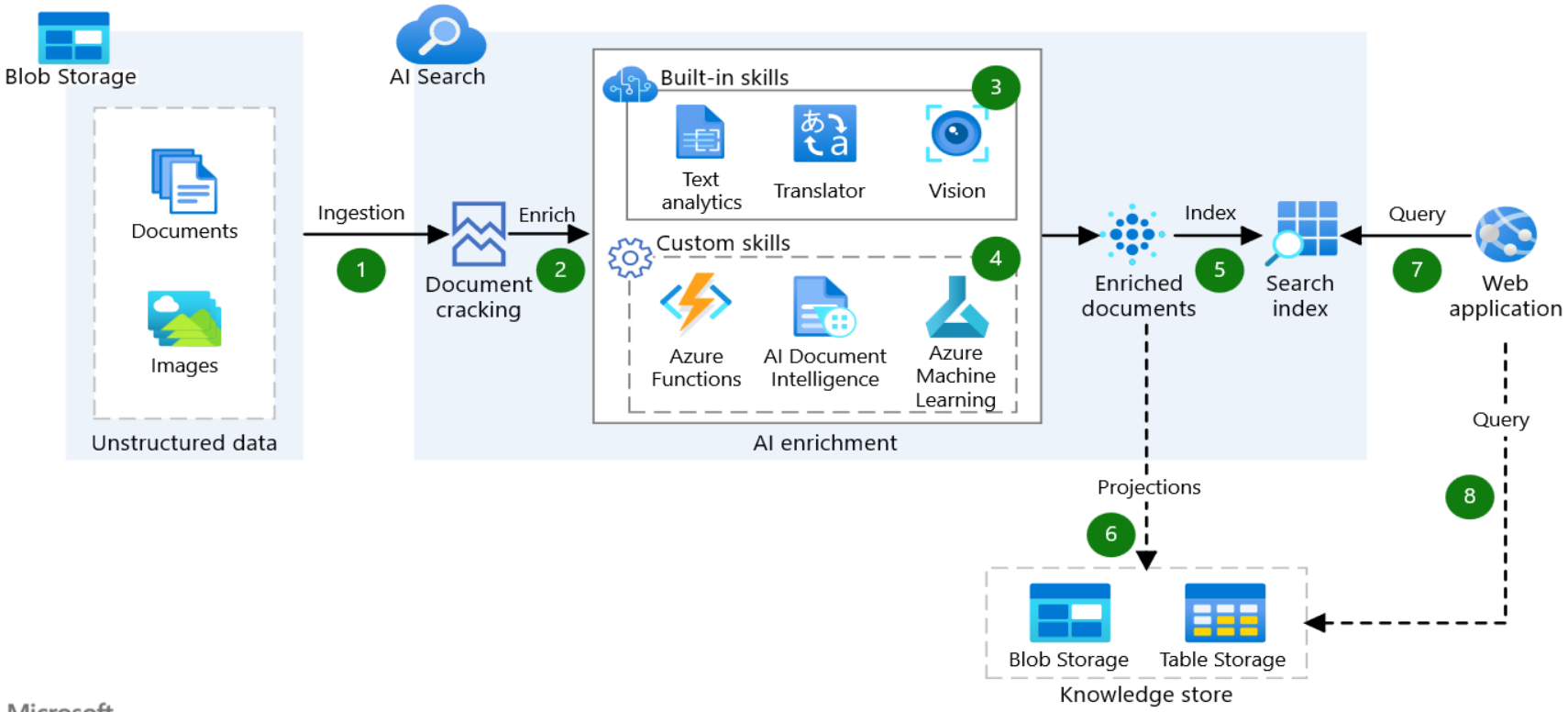
Physical actuators, API calls, generated text, decisions, recommendations

Think of it as a pipeline: Input → Process → Improve → Output

PMT's Architecture



Azur's text and image processing Architecture



Microsoft Azure

Case Study: The Self-Driving Car

Slide 3 of 12 — Real-World Application

The autonomous vehicle is the quintessential AI system — every layer of the architecture is visible in action.

👁️ Perception

LiDAR, cameras & radar detect pedestrians, lane markings, traffic lights, and road signs in real time. ~10M data points/sec.

🧠 Reasoning

Path planning algorithms evaluate 1,000s of possible routes per second, choosing optimal speed, lane & direction.

📚 Learning

Trained on billions of driving miles — both real and simulated — the system continuously improves edge-case handling.

⚙️ Action

Electromechanical systems execute steering, throttle, and braking commands with millisecond precision.

🏆 Industry Context: Companies like Waymo, Tesla & Baidu use this exact architecture, processing terabytes of sensor data daily.

AI in Society: Positive Impacts

Slide 4 of 12 — Benefits & Applications

Healthcare

- Early disease detection (cancer, Alzheimer's) via imaging AI — up to 94% accuracy. Read Harvard article on AI cancer detection
- Drug discovery accelerated from 10 years to under 2 years using generative models. AI drug discovery company Insilico Medicine
- Personalized treatment plans & real-time patient monitoring

Industry & Manufacturing

- Predictive maintenance reduces equipment downtime by up to 50%
- Computer vision quality inspection detects defects at microscale
- Supply chain optimization saves billions in logistics costs

Daily Life & Society

- Voice assistants handle billions of queries daily (Alexa, Siri, Google)
- Recommendation engines drive 35% of Amazon and 80% of Netflix content viewed
- AI tutors provide personalized education to underserved communities globally

$15.7T	Global AI economic impact by 2030
94%	Cancer detection accuracy (PathAI)
50%	Reduction in maint. downtime
80%	Netflix views driven by AI rec. engine

Case Study 1: AI in Healthcare — IBM Watson

Slide 8 of 14 — Applied AI Case Study

🧠 Story: Helping Doctors Make Better Decisions

🔍 What IBM Watson AI Does

- Reads millions of medical research papers in seconds
- Analyzes patient history, medical reports & clinical data
- Cross-references symptoms with latest treatment guidelines
- Flags potential drug interactions & contraindications

📊 Real-World Impact

14+
Countries using IBM Watson Health

70%
Reduction in manual medical coding work

💡 **Outcomes: Faster diagnosis · Better treatment recommendations · Reduced workload on doctors**

Hospitals across oncology departments started using Watson to assist doctors in cancer treatment — combining AI's encyclopedic knowledge of literature with the clinician's judgement.

🔗 **Learn More**
ibm.com/think/topics/artificial-intelligence-medicine

📌 *Watson proved that AI can parse literature faster than any human — a doctor reads ~200 papers/year; Watson reads millions.*

AI in Education — RAG-Based Intelligent Tutor

Slide 9 of 14 — Applied AI Case Study

🧠 Case Study: RAG-Powered AI Teaching Assistant (Higher Education)

📖 Problem

- Students search across PDFs, notes & lectures manually
- Hard to get contextual, accurate answers quickly
- Traditional AI gives generic or hallucinated responses
- No single source of truth across course materials

🚀 RAG Architecture

1. RETRIEVE
Fetch relevant chunks from textbooks, notes & research papers

2. AUGMENT
Inject retrieved context into the LLM prompt for grounding

3. GENERATE
LLM produces accurate, context-grounded answer for student

Retriever + LLM = Accurate Context-Aware Answers

📊 Real Impact

10–15%
Accuracy improvement over standard LLMs

24/7
AI tutor availability for students

Used in chatbots, tutoring & assessment tools

💡 **RAG combines: Retriever (semantic search) + LLM (generation) → Accurate, context-aware, hallucination-reduced answers**

Universities are integrating RAG tutors with LMS platforms (Moodle, Canvas) — students get personalised answers grounded in their actual course material.

AI in Society: Critical Challenges

Slide 5 of 12 — Risks & Responsibilities

💼 Job Displacement

The World Economic Forum estimates AI could displace 85 million jobs by 2025 — but also create 97 million new roles. Net positive, but transition is painful. Reskilling is the policy imperative.

⚖️ Bias & Fairness

Amazon's AI hiring tool discriminated against women. COMPAS recidivism software showed racial bias. AI amplifies existing societal inequalities unless actively counteracted in training data & model design.

🔐 Privacy & Surveillance

Clearview AI scraped 3B+ facial images without consent. Predictive policing raises civil liberties concerns. China's social credit system exemplifies AI-powered mass surveillance at scale.

🤖 Ethics & Transparency

Autonomous weapons ('killer robots') raise international law questions. Black-box AI decisions in lending, healthcare & criminal justice demand explainability. Who is accountable when AI harms someone?

⚠️ Example: Resume-screening AI tools — fast and scalable, but risk encoding historical hiring biases if not audited regularly.

The AI Revolution: Drivers & Forces

Slide 7 of 12 — Why Now?

Big Data + GPUs + Cloud Computing + Better Algorithms = AI Revolution

Big Data

Volume doubled every 2 years (Moore's Law of Data). IoT, social media & mobile created the training fuel AI needed.

GPU Computing

NVIDIA's CUDA enabled parallel processing. Training GPT-4 used ~25,000 A100 GPUs over months. Hardware = enabler.

Cloud Platforms

AWS, Azure & GCP democratized compute. A student today can train a model on hardware that would have cost millions in 2010.

Algorithms

Backprop + Transformers + Attention mechanisms. The Transformer (2017) was the single most important algorithmic leap.

🛒 E-Commerce Example: Amazon's AI recommendation engine drives ~35% of total revenue. It combines collaborative filtering, real-time A/B testing, and demand forecasting into one seamless system.

Real-World Impact Across Sectors

Slide 8 of 12 — Applied AI

Business & Finance

- Algorithmic trading executes millions of orders per millisecond
- AI-powered fraud detection saves $12B+ annually
- LLMs draft legal documents, contracts & financial reports

Automation & Copilots

- GitHub Copilot writes ~40% of code for users who adopt it
- ChatGPT handles customer service at scale, 24/7
- AI process automation (RPA) eliminates repetitive office tasks

Robotics & Manufacturing

- Boston Dynamics Atlas performs complex physical tasks
- Amazon warehouses use 750,000+ robots for logistics
- AI-guided surgical robots improve precision in OR by 40%

AI is not replacing human judgment — it is augmenting it. The best outcomes come from Human + AI collaboration.

Current Trends in AI (2024–2025)

Slide 11 of 14 — State of the Field

01

Generative AI

Text, image, code, audio & video generation. GPT-4o, DALL·E 3, Sora, Gemini 1.5.

02

World Models (Janus)

Unified models that understand AND generate images. DeepSeek Janus-Pro decouples vision encoding from generation for superior multimodal performance.

03

Multimodal AI

Models processing text + image + audio + video simultaneously (GPT-4o, Gemini). Closing the gap between modalities.

04

Foundation Models & RAG

One large model, many tasks. RAG augments LLMs with real-time retrieval — reducing hallucination & improving accuracy.

05

AI Agents & Multi-Agent Systems

Networks of AI agents collaborating on complex tasks: AutoGPT, CrewAI, LangGraph, OpenAI Swarm.

06

AI + IoT + Robotics

Edge AI in smart homes, autonomous vehicles & industrial robots. Boston Dynamics + AI = physical intelligence.

07

Ethical & Regulated AI

EU AI Act (2024), NIST AI RMF. Transparency, fairness & accountability now legally mandated in high-risk systems.

08

Small Language Models (SLMs)

Phi-3, Mistral 7B, Gemma — powerful, efficient models running on-device. Democratising AI beyond the cloud.

🌐 **Key insight: AI is no longer one field — it is a convergence of language, vision, robotics, and reasoning happening simultaneously.**

Future Directions in AI

Slide 10 of 12 — Looking Ahead

Explainable AI (XAI)

Making AI decisions understandable to humans. Regulatory demand is soaring — the EU AI Act mandates explainability for high-risk applications (healthcare, finance, criminal justice).

2030 XAI market: $21B

AI in Education

Adaptive learning systems like Khan Academy's Khanmigo provide 1:1 Socratic tutoring at scale. AI could solve the global teacher shortage while personalizing content to each student's learning style.

EdTech AI: $10B by 2026

AI & Climate Change

DeepMind's GraphCast predicts weather 10 days ahead at higher accuracy than legacy models. AI optimizes power grids, models carbon capture, and designs more efficient materials.

Can cut emissions 4% by 2030

Human–AI Collaboration

The future is not AI replacing humans but AI amplifying human capability. Augmented cognition, AI-assisted creativity & decision support systems will redefine what it means to work.

67% of jobs will be augmented